Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The UDP protocol reveals that: when you send UDP packets to the DNS server, you receive ICMP packets containing the error message udp port 53 unreachable. This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: a network diagnostic message sent by a target device to confirm wasn't reachable The port noted in the error message is used for: Port 53 is a port for DNS service used to find the right server The most likely issue is:your device cannot translate web addresses into IP addresses, making it unreachable

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
Time incident occurred: Between the time of 1:24PM-1:28PM Explain how the IT team became aware of the incident: Several customers of clients reported that they were not able to access the client company website and saw the error "destination port unreachable" after waiting for the page to load. Explain the actions taken by the IT department to investigate the incident: When attempting to access a website, you receive a "Destination Port Unreachable" error. Using `tcpdump`, you discover that DNS queries sent via UDP to port 53 receive ICMP responses stating "UDP Port 53 Unreachable," preventing the browser from resolving the domain name and loading the webpage. Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): The log shows that the computer sent UDP requests to the DNS

server to resolve the IP address of `yummyrecipesforme.com`. The server responded with ICMP error messages indicating that UDP port 53 was unreachable. The timestamps show when the incident occurred, while the source and destination IP addresses identify the communicating devices. The log also displays DNS query details, including the request for an A record. The same delivery error occurred three times because no service was listening on the DNS server's port 53.

Note a likely cause of the incident: the DNS server's service is unavailable or UDP port 53 is blocked